

Celestial Motion in the Quran and Euler Based Orbital Modelling

An interpretive and computational study for learning academic writing

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Abstract

This paper examines how selected Quranic descriptions of celestial movement can be discussed alongside mathematical models of orbital motion. Its purpose is to develop a carefully bounded connection between scriptural interpretation and computation, while demonstrating the structure of an academic paper. The textual corpus consists of Quran 21:33, 36:38–40, and 55:5, read with linguistic resources and selected passages from an abridged English presentation of Ibn Kathir. A focused review considers scholarship on scientific interpretation, cosmography, and the concept of orbit. The computational component compares explicit Euler and symplectic Euler in a dimensionless Newtonian two-body problem, using circular and eccentric initial conditions and an analytical Keplerian reference. Twelve numerical runs cover ten reference periods with 100, 500, or 2,000 steps per period. At 500 steps per period, the circular experiment produces maximum relative energy errors of approximately 44.1% for explicit Euler and 0.0158% for symplectic Euler. The latter preserves angular momentum to floating-point precision, although trajectory and energy errors remain. The interpretive analysis identifies a plausible correspondence at the level of celestial movement and regularity, but the selected passages do not supply the force law, initial conditions, or integration scheme. The paper concludes that Euler-based modelling can illustrate physical motion in dialogue with Quranic reflection. It cannot establish historical influence on Euler, uniquely derive modern mechanics from the verses, or turn numerical accuracy into theological proof.

Keywords: Quran; celestial motion; falak; tafsir; Euler method; orbital mechanics; numerical error; science and religion.

1 Introduction

Connecting a religious text with mathematics requires a clear account of what is being connected. A verse may describe a visible phenomenon, give it theological significance, or invite reflection on it. A mathematical model specifies variables, relationships, and conditions under which a calculation is meaningful. A computer program then approximates the consequences of that model. These activities can inform one another, but they answer different questions. This paper begins with that distinction because an attractive orbital animation can otherwise appear to establish much more than its calculation actually demonstrates.

For a Muslim reader, descriptions of the Sun, Moon, night, and day can encourage attention to the natural world. Studying motion may deepen appreciation of creation and introduce questions about regularity, measurement, and the limits of human knowledge. This motivation is legitimate within a personal learning project. Academic writing asks the writer to take a further step: identify an argument that another reader can evaluate, state its evidence, and explain how the conclusion

follows. Religious commitment can guide the choice of subject without determining every finding in advance.

The present study asks how selected Quranic passages about celestial movement relate to mathematical orbital models, and what role Euler's numerical method can play in explaining that relationship. The question has an interpretive part and a computational part. The interpretive part concerns what the selected language can reasonably be understood to say. The computational part concerns the behaviour of two algorithms applied to an explicitly stated gravitational system. Their combination is educational: it shows how a paper can connect disciplines while keeping the evidence for each claim visible.

Three meanings of support need particular care. A physical model can support an explanation of an observed trajectory when its predictions agree with independent measurements. It can support an illustration of an interpretation when the illustration is consistent with that interpretation. A historical source can support a claim of intellectual influence when it documents a connection between people, texts, or institutions. These are different evidential relationships. A simulated ellipse does not document what Euler read, and a Quranic phrase about movement does not determine a unique numerical algorithm.

Similarly, the word foundation can refer to motivation, interpretation, or mathematical derivation. A believer may regard revelation as a foundation for seeking knowledge. A writer may use a verse as the starting point for a discussion. A mathematical foundation is more specific: assumptions and definitions must permit an equation or theorem to be derived. This paper investigates the first two relationships and explicitly identifies the additional assumptions needed for the third. It does not conduct archival research into Euler's biography or claim that his work originated in the Quran.

Euler is also an ambiguous name in mathematics. Euler's complex exponential identity concerns the relationship between trigonometric functions and complex numbers. Euler–Lagrange equations provide a variational formulation of dynamics. Euler's numerical method advances the approximate solution of a differential equation in discrete steps. The present paper uses the numerical method because it creates a manageable computational example and makes the distinction between a physical law and its numerical approximation especially clear. The broader terminology is discussed so that the title does not imply that all equations associated with Euler describe planetary motion.

The study has four objectives. It identifies the relevant textual statements without replacing their context with modern terminology. It evaluates a small set of related publications. It constructs and reports an orbital experiment that can be reproduced. Finally, it explains which connections the combined analysis supports and which require further evidence. The resulting paper is a personal study exercise with genuine calculations, rather than a claim to have discovered a new physical law or completed a comprehensive study of Quranic interpretation.

2 Review of Related Scholarship

The literature was selected for its direct relevance to interpretation, orbit, or numerical modelling. It is a focused narrative review rather than a systematic review with an exhaustive database search. Consequently, its function is to locate the present argument among existing approaches and identify

useful methodological lessons. The selection should not be read as a ranking of journals or as evidence that all positions in the science-and-religion debate have been represented.

Guessoum's published abstract criticises attempts to identify modern scientific discoveries as uniquely encoded in the Quran and favours readings that can be consistent with scientific knowledge [4]. That position is relevant because it distinguishes interpretive compatibility from proving that a theory is contained in a text. Its value here is methodological: it encourages the writer to make the interpretive step explicit. This study uses the abstract for that limited purpose and does not attribute detailed arguments from the inaccessible article body to the author. A complete critique of his position would require engagement with the full article and its wider philosophical assumptions.

Chauvet examines how Quranic cosmographical language can be read alongside different historical and modern worldviews [5]. His discussion of unspecified orbital centres helps show why agreement with one model does not automatically exclude every alternative. He also develops an analogy between swimming and physical characteristics relevant to orbital behaviour. That analogy is interpretive rather than an equation of mechanics: buoyancy and gravitational motion involve different mechanisms. The paper therefore offers both a useful account of interpretive flexibility and an example of why an evocative physical analogy needs separate mathematical examination. Accepting its broader question does not require accepting every proposed correspondence.

Hoque and colleagues connect Quranic and hadith references with contemporary accounts of orbit and practical astronomical applications [6]. Their article is useful for surveying the theological and observational relevance of celestial movement. Its discussion is predominantly qualitative; it does not provide the controlled Euler comparison undertaken here. Several entries in its bibliography point to general websites or repositories rather than precise article records. That limits the ease of following some supporting claims. The present paper therefore cites the article's own argument but does not inherit its bibliography as verified evidence. A publication can be relevant to a research question while still needing careful source checking.

Azmi and Saadon combine interpretation with computational examples and distinguish galactic solar motion from the Sun's apparent path in an observer's sky [7]. Their modelling uses Python and Gala and reports initial phase-space parameters. This distinction is useful because the phrase solar orbit can otherwise conceal a change of scale or reference frame. Their example operates at a different physical scale from the two-body experiment here. The paper also identifies limits in its selected commentaries and astronomical comparisons. It demonstrates that a computational illustration can accompany interpretive work without presenting the code as a substitute for textual reasoning.

Hernandez and Dehnen analyse numerical methods for planetary systems, including Hamiltonian splitting and symplectic integration [8]. Their introductory treatment provides a technical basis for distinguishing an orbital model from the approximation used to solve it. They describe symplectic Euler through successive momentum and position updates and discuss the accuracy limitations of that simple scheme. Their research concerns more elaborate planetary calculations than the teaching experiment here. Its contribution to this paper is the numerical principle that the geometry and error behaviour of an integrator matter; it supplies no evidence about Quranic interpretation or historical influence on Euler.

These publications leave room for a modest educational contribution. The present exercise makes the assumptions linking text, model, and algorithm explicit, then tests the algorithm against a known

solution. It does not claim that this combination is unprecedented. Its purpose is to show a learner how to separate a literature review from original analysis, how to report reproducible results, and how to avoid transferring the authority of a calculation to a conclusion the calculation did not test.

3 Methodology

The study combines a limited textual analysis with a deterministic numerical experiment. The textual corpus was chosen because the selected passages directly concern the Sun, Moon, movement, or reckoning. Quran 21:33 and 36:40 provide the central references to *falak*. Verses 36:38–39 supply the surrounding solar and lunar context, while 55:5 supplies a related reference to calculation. Other cosmological subjects, such as the origin of the universe, mountains, or the expansion of space, are excluded to keep the argument focused.

The Arabic text and accessible English translation were consulted through Quran.com [1]. The Quranic Arabic Corpus was used as a guide to morphology and vocabulary [2]. Selected passages from the online abridged English Ibn Kathir were examined as one example of exegetical treatment [3]. These resources provide an accessible starting point, but the study does not claim a critical-edition comparison across the entire classical tradition. Modern publications are read as arguments about the text, not as replacements for the text or as independent authorities on the meaning of every Arabic term.

Each passage is examined through four questions. What objects or phenomena are named? What language of movement or regularity appears? What interpretation is being considered? What extra physical information would be necessary to construct a model? The last question is essential. It prevents the interpretation from silently importing mass, gravitational attraction, an orbital centre, or a numerical time step and then presenting those imported assumptions as words contained in the passage itself.

The numerical experiment uses a planar Newtonian relative-motion model. It represents two isolated point masses through their separation vector, with their combined gravitational parameter normalised to one. The semi-major axis is also one. Circular and eccentric cases begin at periapsis with eccentricities zero and 0.5. These choices create a controlled mathematical benchmark. They do not represent measured parameters for the real Earth–Moon system, model the Sun's galactic environment, or reconstruct the entire arrangement of bodies mentioned in the verses.

Explicit Euler and the velocity-first form of symplectic Euler are compared under identical starting conditions. Each method is run for ten analytical reference periods using 100, 500, and 2,000 steps per reference period. The calculations use double-precision NumPy arrays. An analytical Keplerian trajectory, evaluated through Kepler's equation, supplies a comparison independent of the two Euler update rules. The twelve runs therefore differ only in eccentricity, algorithm, and step size, rather than in physical parameters chosen to favour one outcome.

The recorded quantities are maximum relative energy error, final position error divided by the semi-major axis, and maximum relative angular-momentum error. These quantities test different properties. Energy error concerns the computed orbit's energetic state, position error concerns where the body is at a specified time, and angular momentum measures a conserved quantity associated

with the central force. None is treated as a numerical score for a verse. The interpretation is evaluated through textual reasoning; the simulation is evaluated through mathematical diagnostics.

4 Analysis of the Selected Passages

4 1 Movement and Falak

Quran 21:33 names night, day, the Sun, and the Moon before describing movement in a falak. The phrase also appears in 36:40. The Arabic expression *kullun fi falakin yasbahun* is central to this study. Khattab's translation of 21:33 includes the short phrase “each travelling in an orbit” [1]. The English word orbit is useful for a modern reader, but the argument must still return to the Arabic expression and its context. The original does not contain a coordinate system, a specified eccentricity, or a formula for gravitational acceleration.

The Arabic Corpus distinguishes falak from fulk within the same root entry and records falak in the two selected passages [2]. This makes root-based shortcuts unreliable: finding related letters elsewhere does not mean all derived words have the same sense. In the abridged Ibn Kathir passage, the movement is explained through revolving and a circular comparison, with distinct courses and allotted times for the Sun and Moon [3]. This provides an exegetical example of motion and regularity. It does not establish that every commentator understood the phrase identically or that the comparison supplies modern orbital elements.

The present analysis takes celestial movement as the strongest direct theme and treats a modern gravitational orbit as one way of illustrating movement. The inference remains bounded. If an interpreter understands the phrase to describe bodies following their courses, a model displaying a continuous gravitational trajectory can be consistent with that understanding. The model still obtains its mechanics from additional assumptions. The same phrase does not tell the programmer whether to use explicit Euler, symplectic Euler, a higher-order integrator, or an exact analytical solution.

There is also a problem of referents. Night and day are phenomena rather than massive point particles. An interpretation of the collective expression must consider the surrounding syntax and the roles of the named items. A diagram that places labelled particles called night and day on two planetary ellipses would impose a physical ontology without establishing it from the text. This example shows why a visually tidy correspondence may become interpretively misleading even before any mathematics is calculated.

4 2 Solar Movement and Reference Frames

Verse 36:38 introduces solar movement and must be considered alongside the surrounding verses. The abridged Ibn Kathir treatment records interpretations involving an appointed location or course and an appointed end in time [3]. Their presence makes it inappropriate to assume that one modern English formulation exhausts the discussion. A careful argument should identify which interpretation it is using and distinguish that choice from a technical measurement of solar motion.

A further issue is the frame in which a movement is described. An observer on Earth experiences a daily solar path across the sky. An astronomical description can instead locate the Sun within a galactic frame. These descriptions concern different relationships between observer, coordinates, and motion. The statement that the Sun moves does not, by itself, choose between those frames. For the

present exercise, a two-body model is therefore presented as a teaching example, with no claim that it computes the full meaning of 36:38.

This distinction matters when connecting a verse to a figure. The figure needs a caption that states what the axes measure and which body or separation vector is represented. A circular curve drawn on a page could show an apparent path, a physical orbit, or simply a geometric construction. Without that information, the reader cannot tell what has been demonstrated. Academic writing makes the representational choice visible instead of relying on a familiar shape to imply the answer.

4 3 Lunar Stages and Celestial Order

Verse 36:39 concerns the Moon's stages, while 36:40 connects the Sun, Moon, and alternation of night and day within a larger statement of order [1]. This paper treats these as related contextual descriptions rather than independent predictions of a modern algorithm. A claim about order may be compatible with several physical explanations. To select a particular explanation, the researcher needs more than the presence of recurring movement in both the text and the simulation.

A trajectory alone does not model lunar phases. Phase appearance depends on the geometry of illumination and observation. A program can accurately advance a Moon-like point along a curve yet say nothing about the illuminated portion visible to an observer. Connecting the experiment to 36:39 would therefore require an additional light-source and viewing model. Leaving that extension out is a practical way of keeping the present study's claims aligned with what was actually implemented.

Likewise, the wording about the Sun and Moon in 36:40 should not be converted automatically into a universal physical prohibition on astronomical alignment, eclipse, or collision. Those stronger propositions would require their own interpretation and evidence. The numerical experiment does not test them. It supplies an example of regular gravitational motion under specified conditions, while the passage is considered in its own sequence of solar, lunar, and temporal descriptions.

4 4 Reckoning and Mathematical Calculation

Verse 55:5 associates the Sun and Moon with husban, a term connected here with reckoning or calculation [1]. For this study, it provides a natural point of contact with the activity of measuring and computing. The interpretive inference is that celestial phenomena invite discussion of regularity and calculability. The stronger claim that the verse supplies an equation would require the equation's specific relationships to be recoverable from the text, which this exercise does not establish.

A numerical method also has a humanly chosen resolution. Its time step controls how often the program updates its approximate state. The celestial body does not physically advance in visible Euler-sized jumps. Confusing these two kinds of discreteness would mistake a computational technique for a property asserted about creation. The link is more defensible when reckoning motivates careful calculation, including inspection of the calculation's limitations, rather than being treated as a name for one particular algorithm.

5 Mathematical Framework

5.1 From Physical Assumptions to an Orbit

In an isolated Newtonian two-body system, let r be the displacement of one mass from the other. Its relative acceleration is directed toward zero separation and has magnitude proportional to the inverse square of the distance. The combined parameter is $\mu = G(M + m)$, where G is the gravitational constant and M and m are the two masses. The standard relative-motion equation is given in the classical astrodynamics material used here [9]. The model is written as two first-order equations so that an integration method can advance position and velocity.

$$\dot{r} = v, \quad \dot{v} = a(r) = -\mu \frac{r}{\|r\|^3} \quad (1)$$

The distinction between equation and solution is useful. Equation 1 defines how velocity changes at each position. To determine a particular trajectory, the problem also needs an initial position and velocity. Different initial velocities can lead to different trajectories under the same acceleration law. A curve therefore cannot be inferred from a statement of motion alone; it is the consequence of a model together with its initial state.

Dimensionless variables simplify the experiment. Taking the semi-major axis a and the gravitational parameter μ as units gives a reference time scale equal to the square root of a^3/μ . With $a = \mu = 1$, the chosen bound reference orbit has period $T = 2\pi$. The system's numerical values then concern a representative mathematical orbit. Restoring physical units would require actual masses and distances, which the present benchmark intentionally does not assign to scriptural terms.

5.2 Euler Updates

Explicit Euler follows from approximating a smooth solution by its tangent over a short time step h . For a general differential equation $\dot{y} = f(y, t)$, the next value is the current value plus h times the current derivative. Applying that reasoning separately to the two parts of Equation 1 gives the following scheme. Both updates use the state at the beginning of the step, so an implementation must preserve the old values until both calculations are complete.

$$r_{n+1} = r_n + h v_n, \quad v_{n+1} = v_n + h a(r_n) \quad (2)$$

The velocity-first symplectic variant changes the order of dependence. It first computes the velocity and then uses that new velocity to update position. This is the elementary kick-and-drift structure discussed by Hernandez and Dehnen [8]. The formula is simple enough to inspect directly, making it useful for a learner who wants to understand how a small implementation choice can affect the shape of a simulated orbit.

$$v_{n+1} = v_n + h a(r_n), \quad r_{n+1} = r_n + h v_{n+1} \quad (3)$$

Both methods are first order in general. The omitted Taylor terms produce a local error of order h^2 ; under the usual smoothness and bounded-interval conditions, accumulated global error is generally of order h . This classification does not guarantee the same energy behaviour, stability, or

cost over many periods. Nor does it prevent specially chosen initial conditions or measurement times from showing a faster decrease in a particular diagnostic. A paper should distinguish the general method order from one favourable error pattern.

5.3 Conservation Diagnostics

For the model in Equation 1, define specific relative energy E and the planar angular momentum L . Their initial values provide reference quantities for the experiment. They are computed from the simulated positions and velocities rather than fitted to the expected orbit. This creates a check that can expose inaccuracies even when a plotted curve looks smooth.

$$E = \frac{1}{2} \|v\|^2 - \frac{\mu}{\|r\|}, \quad L = x v_y - y v_x \quad (4)$$

Angular-momentum preservation in Equation 3 can be verified algebraically. The cross product of the new position with the new velocity equals the cross product of the old position with that new velocity, because a vector crossed with itself is zero. The velocity change contributes a cross product of position with acceleration. Central acceleration is parallel to position, so that contribution vanishes. Thus the update preserves angular momentum in exact arithmetic. The small nonzero errors reported below are consistent with floating-point rounding.

Explicit Euler does not have the same property. Expanding its new cross product leaves an additional term involving h squared times velocity crossed with acceleration. Repeated contributions can accumulate. This derivation explains why changing the position update can improve a conserved quantity even though both methods use the same gravitational acceleration. It is a mathematical explanation obtained from the stated algorithm, with no need to assign a special meaning to a scriptural word.

6 Numerical Experiment

The initial separation is placed on the positive x axis at periapsis. For eccentricity e , the starting position is $(1 - e, 0)$ and the starting velocity is $(0, \text{square root of } (1 + e)/(1 - e))$. These conditions give semi-major axis one and initial energy minus one half in both cases. The initial angular momentum is the square root of $1 - e$ squared. Choosing two eccentricities tests whether conclusions obtained from a circle persist when distance and speed vary substantially within a period.

The reference position is evaluated using the eccentric anomaly u . For each sample time, the mean anomaly is reduced modulo 2π and Kepler's equation is solved by Newton iteration. Iteration stops when the maximum correction is below 10 to the power minus fourteen, or after thirty iterations. The experiment checks this reference as a solution of the specified idealised problem; it does not treat it as astronomical observational data.

$$u - e \sin u = t \pmod{2\pi}, \quad x_{ref} = \cos u - e, \quad y_{ref} = \sqrt{1 - e^2} \sin u \quad (5)$$

The step size is 2π divided by the selected number of steps per period. Each run uses ten times that number of steps, ending at ten reference periods. A fixed analytical duration matters because a distorted numerical orbit may take a different time to complete one apparent revolution. Counting ten

revolutions of each computed trajectory would compare different physical times and hide part of the algorithm's error.

The maximum relative energy error is the largest absolute deviation from initial energy divided by the magnitude of initial energy. The position diagnostic is the Euclidean difference from the reference at the final time, divided by a . The angular-momentum diagnostic is defined analogously to energy. All maxima include the initial sample and every later sample. Percentages in the tables multiply the dimensionless relative energy values by one hundred; position errors remain dimensionless multiples of the semi-major axis.

No random sampling enters this experiment. The attached Python script records its Python and NumPy versions, writes the full numerical results to CSV and JSON, and creates the comparison figure. Reproducibility here means that another reader can use the same model, parameters, update rules, and measurement definitions. Small variations in the final rounding digits do not change the reported behaviour or the interpretive conclusions.

7 Results

Table 1 summarises the circular experiment. At 500 steps per reference period, explicit Euler reaches a maximum relative energy error of 44.1251%, while symplectic Euler reaches 0.0157920%. Their final position errors are approximately 1.30172 a and 0.00702814 a respectively. The difference is substantial even though both algorithms evaluate the same acceleration once per step. The outward drift visible in the figure is therefore a numerical effect of the approximation, rather than a change in the physical law.

Table 1 Results for eccentricity 0 over ten reference periods

Method	Steps per period	Max energy error %	Final position error / a
Explicit Euler	100	66.8613	2.78335
Symplectic Euler	100	0.395167	0.175442
Explicit Euler	500	44.1251	1.30172
Symplectic Euler	500	0.015792	0.00702814
Explicit Euler	2000	22.9262	0.575172
Symplectic Euler	2000	0.000986963	0.000439256

Reducing the step size improves both methods across the tested runs. Nevertheless, explicit Euler still has a 22.9262% maximum relative energy error at 2,000 steps per period over the specified duration. A larger number of steps alone is therefore insufficient evidence that a result is accurate enough for a particular purpose. Its adequacy must be judged against a diagnostic and an intended tolerance. The circular symplectic result at the same resolution has a maximum energy error below 0.001%, but that observation is specific to these conditions.

The eccentric experiment, summarised in Table 2, is more demanding. At 500 steps per period, explicit Euler produces a maximum relative energy error of 67.6625% and a final position error of

approximately 5.84828 a. Symplectic Euler gives 1.88191% and 0.172761 a. Its improvement remains clear, but the positional discrepancy is much larger than in the circular run. This difference illustrates why a single visually attractive circular example cannot establish performance for every orbit.

Table 2 Results for eccentricity 0.5 over ten reference periods

Method	Steps per period	Max energy error %	Final position error / a
Explicit Euler	100	114.242	11.6188
Symplectic Euler	100	12.3867	1.99519
Explicit Euler	500	67.6625	5.84828
Symplectic Euler	500	1.88191	0.172761
Explicit Euler	2000	41.2172	3.26216
Symplectic Euler	2000	0.445791	0.0108677

At the coarsest eccentric resolution, explicit Euler's energy error exceeds 100%. Inspection of the calculated energy shows that a system initially specified as bound becomes numerically unbound. This is a failure of the tested numerical approximation under those conditions. Symplectic Euler at that resolution still has a final position error of approximately 1.99519 a. Preserving one invariant therefore does not make a coarse trajectory acceptable, and an algorithm with better structural properties still requires a suitable time step.

Across all twelve runs, symplectic Euler's relative angular-momentum error remains below 1.5 times ten to the power minus fourteen. Explicit Euler's corresponding errors range from approximately 13.9% to 70.0%. This is consistent with the algebraic distinction established above. However, conservation is only one diagnostic. A simulation can preserve angular momentum while advancing its orbital phase inaccurately, and its energy can oscillate while position error develops. The final position metric is included precisely to avoid declaring success on the basis of conservation alone.

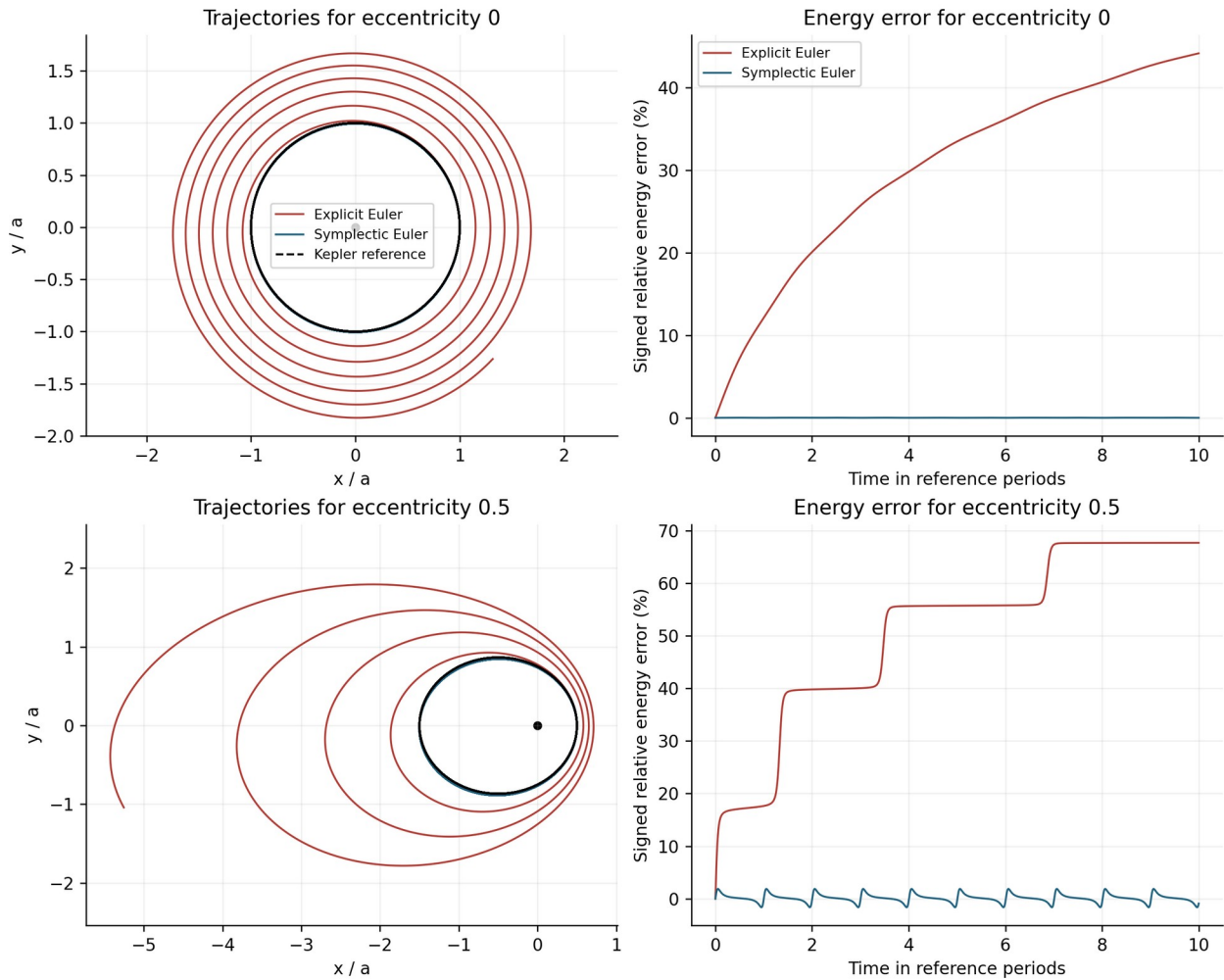


Figure 1 Trajectories and signed relative energy errors for ten reference periods with 500 steps per period. The black point is the relative-coordinate origin. The dashed curve is the Kepler reference.

The figure compares trajectories and signed energy errors at 500 steps per period. Signed error indicates the direction of the energy change, whereas the tables report the largest absolute relative error. The benchmark and its circular case produce some especially favourable symplectic error reductions. These observations should not be promoted into a claim that the general method is second order. They describe these initial conditions, this interval, and these particular diagnostics.

8 Discussion

8.1 What the Comparison Supports

The combined analysis supports a limited educational connection. The passages provide language about celestial movement and reckoning that a Muslim reader can use as a starting point for studying astronomy. The model provides a precise example of motion that can be calculated. The numerical comparison then shows how human approximations succeed or fail when representing that motion. Each part contributes something different: a question of interpretation, a physical hypothesis, or an algorithmic result.

Calling the connection support is reasonable only when the supported proposition is stated. The experiment supports the proposition that, in the tested benchmark, symplectic Euler preserves angular momentum and gives smaller selected errors than explicit Euler. The textual discussion supports the observation that movement and reckoning are themes of the selected passages. Their conjunction supports using the experiment as an illustration in a discussion of those themes. It does not establish that the verses prescribe the successful algorithm or that the less successful algorithm is religiously inappropriate.

This qualification also helps preserve the value of the learning exercise. If a reader's appreciation depends entirely on one algorithm producing an undistorted curve, a coding error or a coarse time step could be mistaken for a problem with the text. Once the computational layer is identified, the error has an appropriate explanation and remedy: inspect the update, reduce the step size, compare another method, or refine the physical model. The meaning of a verse is not measured by a programmer's selection of h .

8 2 Foundation and Historical Influence

The Quran can function as a foundation for this personal inquiry in a motivational sense. The selected passages supply the subject that the learner chooses to examine. The learner then acquires mathematical tools to investigate a physical example and practises explaining the reasoning. This relationship does not require a claim that revelation contains source code or that a later mathematician consciously derived his method from scripture. It is an account of how a present reader connects reflection with study.

Historical influence would be a separate research question. It would require dated evidence concerning access to texts, reading, correspondence, citations, or a traceable chain of transmission. The existence of a Quranic passage before Euler's lifetime establishes chronology, but chronology alone cannot establish intellectual dependence. Many earlier descriptions of the sky preceded later mathematics. To attribute a specific contribution, a researcher must identify the relevant pathway rather than infer it from similarity in subject matter.

Mathematical derivation has another evidential requirement. To derive the Euler step, the learner assumes a differential equation and approximates its solution using a short-time Taylor expansion. To derive the orbital acceleration, the learner assumes Newtonian gravitational dynamics. These are explicit premises in the calculation. Removing them leaves no unique algorithm available from the selected passages. Recognising where the premises enter gives the paper a clearer argument than simply attaching a formula to a verse and describing the pairing as a derivation.

8 3 The Role of Interpretation

Compatibility is a useful finding, but it is logically weaker than unique prediction. If a statement permits several physical interpretations, one compatible model does not establish that the statement selected that model in advance. To argue for uniqueness, a researcher would need to identify discriminating details and show how they rule out alternatives without borrowing those details from later knowledge. The current textual analysis does not perform that stronger demonstration.

Interpretive flexibility need not make the exercise meaningless. It can be the subject of analysis itself. A careful paper can ask how a term has been translated, how an interpretation uses surrounding

context, or how a modern reader brings a physical concept to the text. These are questions that can produce informative answers even when no new astronomical prediction results. They also give the writer material for a discussion section that develops the argument rather than repeating the simulation table.

The present exercise uses a small source selection, so its interpretive conclusions remain correspondingly narrow. It does not settle the permissible boundaries of scientific tafsir for every scholarly tradition. It shows one way of writing about a connection while preserving the distinction between a directly examined passage, a commentator's interpretation, and the writer's own inference. That distinction is especially valuable when a paper addresses readers with different theological commitments.

8 4 Visualisation and Evidence

An animation can be persuasive because smooth movement resembles what a viewer expects an orbit to look like. Yet a smooth curve can be generated by a geometric parameterisation with no gravitational integration at all. A numerical trajectory can also look regular over a short interval while its energy or phase error accumulates. The relevant evidence is therefore the generating method together with its checks, not simply the appearance of the picture.

For an educational orbit project, a useful interface would identify the model, initial conditions, method, and time step. It could let the learner switch between the two Euler updates and inspect energy error alongside the moving body. Any Quranic reference should be accompanied by the specific interpretation being illustrated. This makes the visualisation a transparent teaching tool and gives the learner a concrete reason to connect a figure, a methodology section, and a result paragraph.

9 Limitations and Further Work

The textual component is restricted to a few passages and one abridged online presentation of a classical commentary. It does not establish a comprehensive history of interpretation, independently assess manuscript variants, or provide a full lexical study from critical Arabic dictionaries. An expanded study should compare additional commentaries in named editions and obtain informed Arabic review. Statements about al-Tabari, al-Qurtubi, or other authors should be added only after their relevant passages are actually examined.

The literature review is selective and includes one paper at abstract level. It is adequate for a bounded personal exercise but cannot establish an exhaustive research gap. The reference list records the works used directly, rather than copying unverified citations from their bibliographies. Further reading should test the current argument against contrasting positions and verify sources before expanding the historical discussion.

The model excludes solar perturbations, relativistic corrections, finite body size, tides, and non-gravitational forces. Its results describe a dimensionless isolated two-body benchmark, not a calibrated celestial ephemeris. Only two eccentricities, three resolutions, two algorithms, and one duration are examined. Additional methods, longer intervals, noninteger-period comparisons, and a systematic convergence study could test how broadly the observed behaviour persists.

The experiment also does not evaluate learning outcomes. It shows a proposed educational connection, but no students were recruited and no comprehension scores were measured. A later educational study could test whether displaying numerical error helps learners distinguish physical models from their approximations. Such a study would require an appropriate design and, where applicable, human-participant review. It would create evidence for teaching effectiveness that the present paper cannot claim.

10 Conclusion

The selected Quranic passages can be discussed alongside orbital mathematics through a clearly defined relationship of interpretation and illustration. Their language directs attention to celestial movement and reckoning, while the numerical example shows how a specified gravitational model can be approximated and checked. The connection is most defensible when the writer states the physical assumptions instead of presenting them as hidden content extracted from the verses.

The computational results show that the choice of numerical method changes what the reader sees. In the circular benchmark at 500 steps per reference period, explicit Euler produces a maximum relative energy error of approximately 44.1%, whereas symplectic Euler produces approximately 0.0158%. The eccentric results are less favourable for both methods, and angular-momentum preservation does not eliminate position error. These findings justify careful error reporting when an orbital visualisation is used as part of a textual discussion.

For a learner, the paper demonstrates how to turn a broad interest into a bounded question, use accessible scholarship critically, define a reproducible method, report actual results, and state the limits of the conclusion. It offers a connection between Quranic reflection and Euler-based computation without claiming an unestablished historical dependency or a mathematical proof of religious meaning. The next stage is to deepen the textual comparison and refine the numerical study while preserving that clarity about what each kind of evidence can establish.

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Appendix A Source Review Record

The record below identifies the depth of consultation. Full-text access does not mean that every claim in a publication has been independently verified. The numerical analysis in this paper is a separate calculation and does not reproduce the larger experiments in the reviewed studies.

Publication	Consultation	Role in this paper
Guessoum 2008	Abstract and reference list	Limited account of the published position
Chauvet 2023	Full text accessible through web PDF; focused reading of relevant argument	Interpretation and competing cosmographies

Publication	Consultation	Role in this paper
Hoque and colleagues 2026	Full-text PDF read	Qualitative orbit discussion and source checks
Azmi and Saadon 2026	Full-text PDF read; Malay body and English abstract	Reference frames and computational scope
Hernandez and Dehnen 2017	Full PDF obtained; relevant introductory and method sections examined	Numerical integration and symplectic Euler

The Azmi and Saadon article page gives a publication date of 30 September 2026, while the PDF's front matter gives 20 September 2026. The reference uses the unambiguous publication year. The journal's stated page range is retained even though the PDF includes a preliminary page numbered 53. This discrepancy does not affect the claims used in the present discussion.

Appendix B Reproducing the Numerical Example

Run `orbit_experiment.py` with Python 3, NumPy, and Matplotlib. The script creates `paper_assets/orbit_results.csv`, `paper_assets/results.json`, and `paper_assets/orbit_comparison.png`. The reference mean anomaly equals dimensionless time because a and μ are both one. No measured astronomical dataset is required. The CSV contains unrounded values for all twelve runs, including angular-momentum errors omitted from the compact tables.

The calculation checks should include the initial energy of -0.5 , initial angular momentum equal to the square root of $1 - e^2$, and reference positions equal to the starting periapsis after each full analytical period. The central-force update's angular-momentum identity provides an additional independent check. When adapting the script, change the physical model, integration method, and measurement interval deliberately and record those choices before comparing outcomes.